

MANGILSAN, Korea, Republic of

WMO: 471260

Lat: 36.937N

Lon: 126.444E

Elev: 656

StdP: 14.35

Time Zone: 9.00 (E09)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.2	17.5	3.9	6.9	17.8	6.7	8.0	20.0	28.3	27.6	25.1	28.7	11.3	340	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	86.4	76.9	84.3	75.9	82.3	74.7	78.4	84.2	77.3	82.5	76.1	80.7	8.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	143.8	82.2	75.4	136.5	80.3	74.8	133.6	79.6	42.6	84.7	41.3	81.8	40.0	82.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.3	24.4	20.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	29.4	32.6	39.9	50.6	60.8	68.6	75.2	77.1	69.8	59.6	47.2	34.5
	DBStd	17.35	6.52	6.72	5.43	6.44	5.10	4.39	4.24	3.88	4.45	5.35	7.38	7.59
	HDD50	2133	637	486	315	68	1	0	0	0	0	6	138	482
	HDD65	5052	1102	906	779	433	148	16	0	0	10	177	535	946
	CDD50	3552	0	0	1	87	336	559	780	839	593	303	53	1
	CDD65	998	0	0	0	2	19	125	315	374	154	9	0	0
	CDH74	5105	0	0	0	9	46	391	1698	2502	449	10	0	0
	CDH80	1045	0	0	0	2	1	53	355	591	43	0	0	0
Wind	WSAvg	9.5	9.2	10.0	10.9	10.8	9.3	7.8	8.8	8.1	8.6	9.4	10.5	10.4
Precipitation	PrecAvg	50.4	0.9	1.3	1.8	3.1	3.8	6.1	11.1	10.9	5.8	2.1	2.0	1.3
	PrecMax	70.6	2.6	3.3	4.9	7.5	8.4	14.3	26.1	35.7	15.7	7.1	6.3	2.5
	PrecMin	31.8	0.2	0.1	0.4	0.9	0.7	1.1	4.4	1.1	0.5	0.1	0.4	0.2
	PrecStd	10.5	0.6	0.9	1.1	1.6	1.8	3.5	4.7	7.7	4.4	1.9	1.4	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	55.8	59.3	73.2	78.6	84.4	91.1	90.6	84.2	75.3	66.2	55.4
		MCWB	43.9	48.4	50.2	60.1	64.0	69.9	78.0	77.8	74.3	63.2	58.8	48.7
	2%	DB	43.0	48.4	54.3	67.7	74.8	80.9	86.2	87.7	80.9	71.8	62.7	50.2
		MCWB	39.0	43.5	46.8	56.5	61.9	68.8	77.2	77.6	71.6	61.8	56.3	45.1
	5%	DB	40.9	44.6	51.6	64.3	71.7	78.5	83.9	85.7	78.7	69.8	60.2	46.7
		MCWB	37.3	40.6	45.1	54.3	60.6	68.1	76.2	77.1	70.3	61.1	54.1	43.0
	10%	DB	39.2	42.4	48.4	60.7	68.4	75.5	80.9	83.4	76.8	67.8	57.2	44.6
		MCWB	36.0	38.8	42.8	51.7	59.4	66.6	74.7	75.9	69.3	59.2	52.2	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	50.4	51.8	62.1	67.3	73.8	80.1	80.2	76.6	66.8	61.2	50.3
		MCDB	49.1	51.6	57.6	70.1	73.7	78.6	86.5	86.2	81.3	72.2	64.8	55.0
	2%	WB	39.6	44.1	48.3	58.3	64.5	71.6	78.3	78.8	74.3	63.9	57.5	46.3
		MCDB	42.0	46.5	54.0	64.8	70.1	76.4	83.9	84.9	78.5	69.4	60.3	49.2
	5%	WB	37.7	41.2	45.3	55.3	62.6	70.0	77.1	77.8	72.4	62.2	55.0	43.4
		MCDB	40.2	43.7	50.3	62.0	69.1	75.2	82.1	83.4	76.5	67.6	59.0	46.5
	10%	WB	35.9	38.5	43.1	53.0	60.8	68.3	75.8	76.8	70.4	60.5	52.4	40.6
		MCDB	38.9	42.0	47.6	59.3	67.2	73.4	79.9	82.0	74.9	65.7	56.6	43.8
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.2	12.6	14.6	13.5	12.2	9.3	10.4	11.2	12.7	11.7	10.9
		MCDBR	13.5	14.7	17.7	19.0	17.1	15.9	12.7	12.8	13.2	14.9	13.7	13.6
		MCWBR	11.7	11.5	13.1	12.1	10.7	9.8	7.2	6.8	7.5	9.1	9.7	11.7
	5% WB	MCDBR	12.3	13.5	16.8	17.5	15.4	13.7	10.6	11.5	11.3	12.9	12.2	13.4
		MCWBR	11.4	11.6	13.1	12.0	10.9	9.1	6.8	7.1	7.7	9.2	9.9	11.9
Clear-Sky Solar Irradiance	taub		0.429	0.473	0.547	0.580	0.599	0.668	0.633	0.558	0.493	0.482	0.465	0.429
	taud		1.975	1.897	1.758	1.732	1.749	1.672	1.803	1.991	2.094	1.984	1.970	1.988
	Ebn at Noon		226	232	227	229	228	212	219	233	241	227	212	215
	Edh at Noon		43	53	67	72	72	78	68	55	47	48	43	40
All-Sky Solar Radiation	RadAvg		772	1033	1393	1618	1795	1664	1365	1445	1346	1160	778	670
	RadStd		50	88	135	115	176	181	208	208	139	86	82	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+95

Nomenclature: See separate page